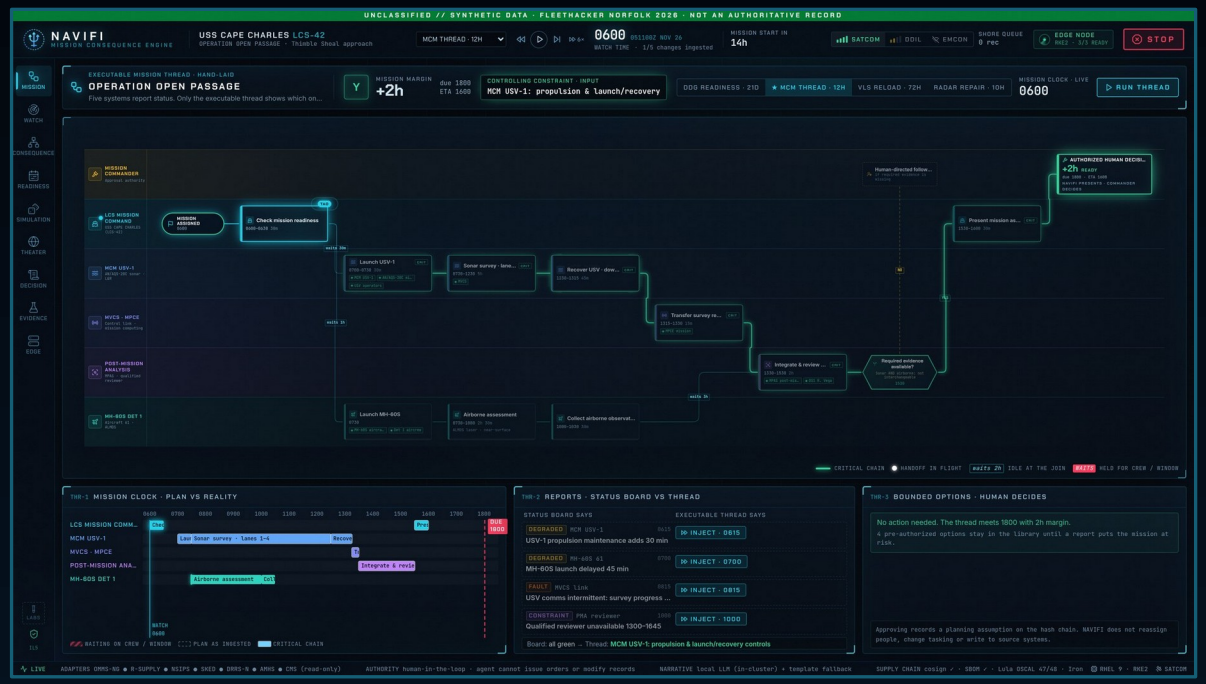
FLEETHACKER NORFOLK 2026

HAWSIR

A mission consequence engine for the watch floor

Know before it breaks the mission.

We built it this week: the app, the engine under it, and a real cluster to run it on. This deck is what we did, what we hit, and where it stands.



TUESDAY

Engine and watch floor

Consequence graph, Monte Carlo, hash-chained decision log, evidence pack, three scenarios

WEDNESDAY

The mission thread

Executable MCM thread, auto-laid threads, Connor's standalone mission engine, the video

THURSDAY

Live on our own cluster

RHEL 9, RKE2 and UDS Core stood up on AWS; HAWSIR deployed from a git commit

WHO WE ARE

A two-day sprint

Questions, or a mission you want to see run through it? Email either of us.



Bryce Schumacher

Built the HAWSIR app: the watch floor, the mission thread, the decision log and the evidence screens. Stood up the lab cluster on AWS.

bschumacher@uncomn.com



Connor Barnsley

Built the standalone mission engine: the dependency graph, Monte Carlo, the decision ledger, and naval and airlift mission packs, with 48 tests.

cbarnsley@uncomn.com

WHERE WE STARTED

Four reports land. Which one breaks the mission?

Today a Sailor answers that by hand. We wanted the watch to see it on one screen.

How it gets answered today

Pull the CASREP
↓
Call supply
↓
Check the watchbill
↓
Redo the whiteboard

30–60 min

per change, by our own tabletop estimate. Not measured fleet data.

What we built instead

HAWSIR reads the records of eight systems, read-only, and recomputes the mission in milliseconds.

Every number links back to the record it came from. Nothing is typed twice or written back.



OMMS-NG | R-SUPPLY | NSIPS | SKED

DRRS-N | WATCHBILL | MVCS | AMHS

Eight systems in one picture. Synthetic records for now.

FIRST, THREE TERMS

MBSE, mission engineering and mission readiness

Engineers model systems and missions. Ships report readiness. Today a Sailor joins the two by hand.

MBSE

Model-based systems engineering

Design a system as one connected model instead of a stack of documents. Requirements, parts, interfaces and behavior link to each other, so a change shows where it lands.

ANSWERS
How is this system built, and what does it need to work?

WHERE IT LIVES
Design and acquisition, usually in SysML

IN OUR DEMO
What USV-1, the MH-60S and the analyst each need: parts, maintenance, a qualified person

→

Mission engineering

Start from the mission, not the platform

Break a mission into a thread of tasks, systems, people and timing, and engineer across all of them so the mission works, not just each system.

ANSWERS
What has to happen, in what order, for this mission to succeed?

WHERE IT LIVES
Planning and analysis; the DoD Mission Engineering Guide

IN OUR DEMO
Operation Open Passage: USV sonar and a helo survey in parallel, an analyst reviews both, report by 1800

→

Mission readiness

Can we do it, and when?

Whether a unit can do its assigned missions, now and when they are needed. The Navy reports it in DRRS-N against mission essential tasks, rated Y, Q or N.

ANSWERS
Will this mission happen on time, and if not, why?

WHERE IT LIVES
DRRS-N, CASREPs, supply status, the watchbill

IN OUR DEMO
MCM rated N: 1 h 15 m short, held up by the analyst's watch

THE DIGITAL THREAD

How mission engineering threads into readiness

Each link points to the next, so a status change finds the mission it hits, and the answer traces back.

BUILT DOWN THE THREAD

1 · MISSION

Mine assessment

Thimble Shoal approach, due to the commander by 1800

From the tasking

2 · MISSION THREAD

Six lanes of tasks

Survey, review and report, with every handoff and its timing

From mission engineering

3 · SYSTEMS AND PEOPLE

Its dependencies

USV-1, the MH-60S and the analyst, and the parts, upkeep or quals each needs

From the system models

4 · LIVE RECORDS

Today's status

CASREP, supply ETA and watchbill, read-only, as they change

From the ship

5 · READINESS

MCM: N, −1 h 15 m

Controlled by OS1 Vega, with bounded options to recover

Y, Q or N, as DRRS-N rates it

Traced back: every answer points to the task, the system and the record behind it

TODAY

The thread is a plan or a slide. Readiness is reported system by system. The watch joins them on a whiteboard, by hand, every time a report lands.

WITH HAWSIR

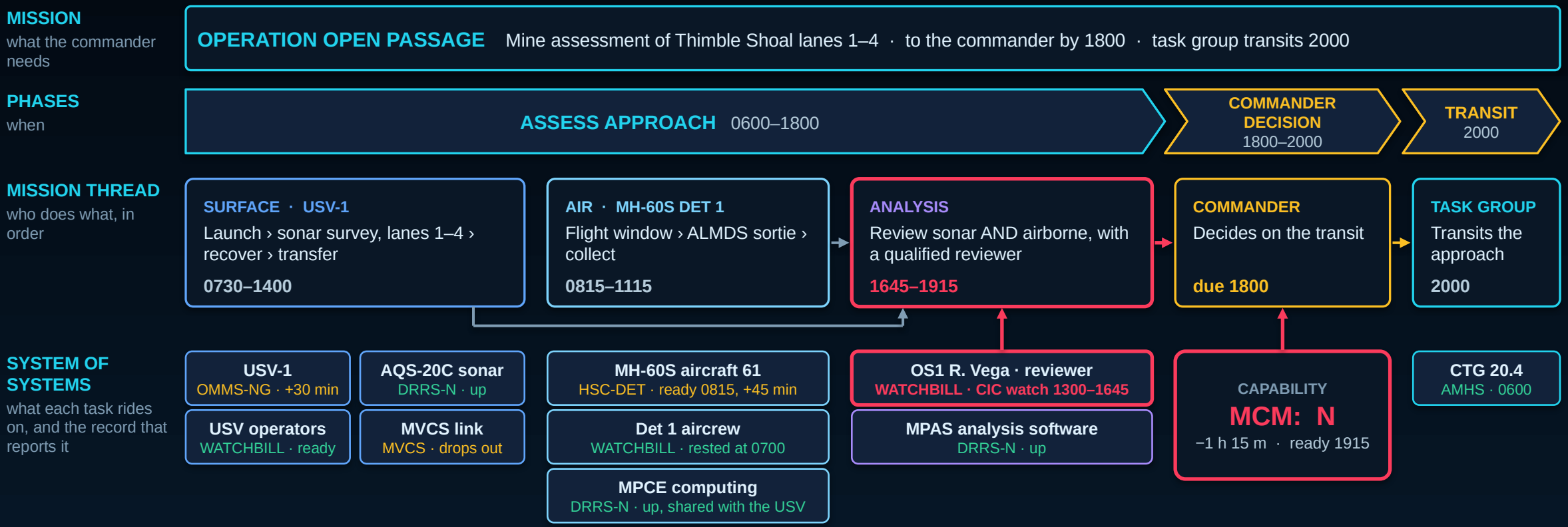
The thread runs. Records attach to its nodes as they change, and readiness is computed per mission: Y, Q or N, the margin, and what controls it.

Not yet: we write the thread and the system dependencies by hand. Reading them from the engineering model is the next link.

MISSION ENGINEERING, CONCRETELY

One mission, from its phases down to the systems it rides on

Operation Open Passage as the engine sees it at 1000. The red path is what controls the mission.

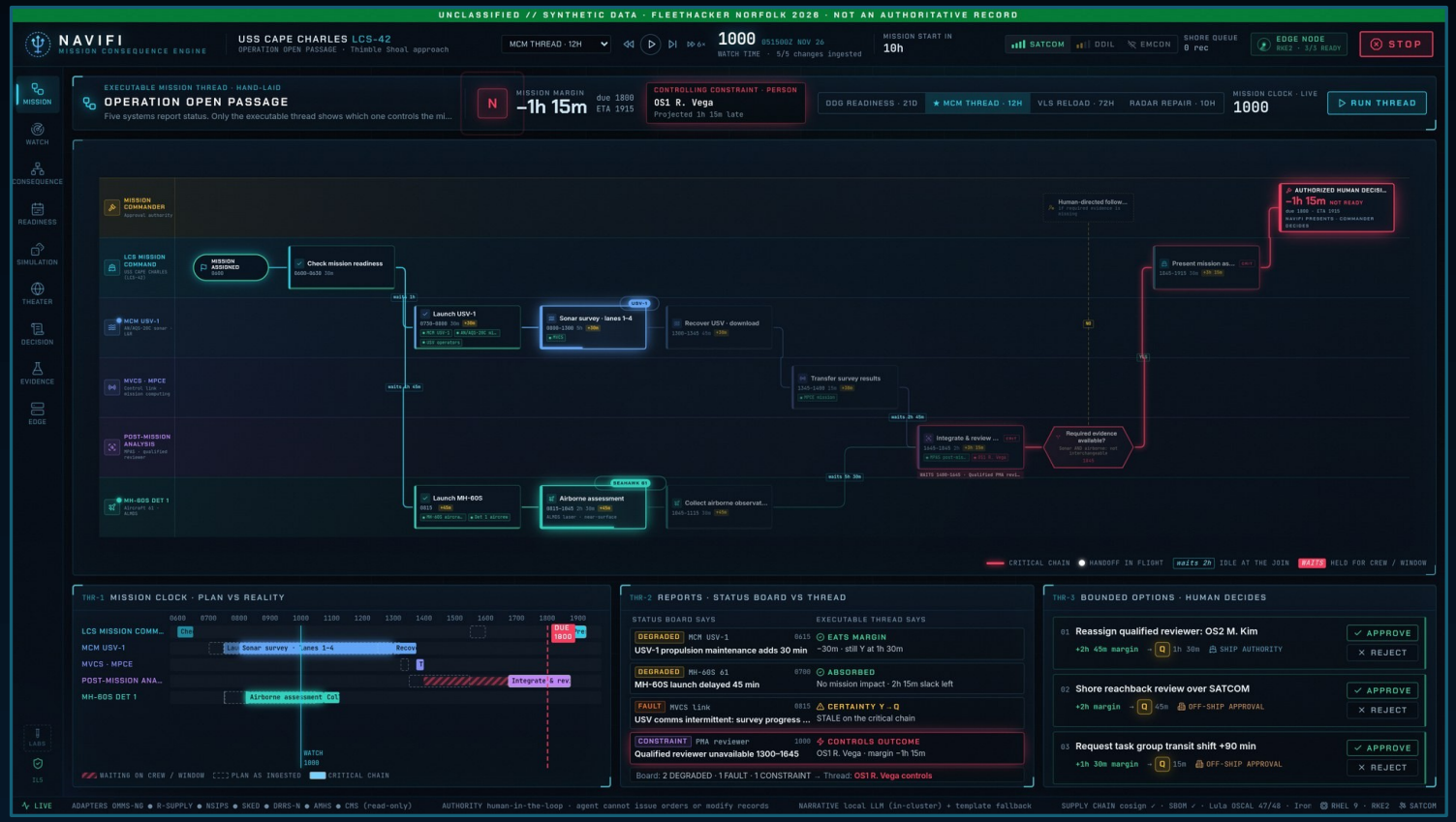


READ IT BOTTOM-UP OS1 Vega takes the CIC watch 1300–1645 on the watchbill › the review can't start until 1645 › MCM is ready at 1915, 1 h 15 m past due › the commander's decision and the 2000 transit slip with it. The USV and helo reports landed first and looked bigger; neither one controls.

WHAT WE BUILT

We made the mission itself executable

Operation Open Passage: mine countermeasures for USS CAPE CHARLES (LCS-42), synthetic data



The live app at 1000, mid-mission

1 Six swimlanes, one thread

Mission commander to the MH-60S detachment, recomputed on every status report.

2 Four reports in. One controls.

At 1000 it points at the analyst's review window, not the degraded USV or helo: −1h 15m, not ready.

3 Fixes simulated, not guessed

Three bounded options run on the same thread. Reassigning a qualified reviewer gets to +1h 30m.

BEFORE AND AFTER

What it gives back to the watch

Our tabletop comparison for one change landing mid-watch. Not measured fleet data yet.

AFTER A CHANGE LANDS	TODAY	WITH HAWSIR	HOW
Time to a defensible answer	30–60 min	**Milliseconds**	Recomputed the moment the report lands
Systems consulted	5–7, plus calls	**One screen**	Every number links to its source record
Nonlinear effects caught	Often missed	**2 of 2**	A missed sail window and a helo fly-on cliff
Data problems surfaced	When they bite	**4 of 4**	Missing, stale, contradictory, malformed
Decision record	Notebook or email	**Hash-chained log**	Who decided what, on which facts, and when

How we would make it real In the first 30 days: sanitized, read-only exports from one DDG, and three watch teams run this same comparison against today's process.

WHERE THE DATA COMES FROM

Nothing typed twice

HAWSIR reads the records the ship already keeps. Nobody re-enters a status to make the picture move.

Read-only connectors

- **OMMS-NG** maintenance
- **R-SUPPLY** parts and ETAs
- **NSIPS** personnel
- **WATCHBILL** watches and quals
- **SKED** PMS schedule
- **DRRS-N** readiness
- **MVCS** unmanned vehicles
- **AMHS** message traffic

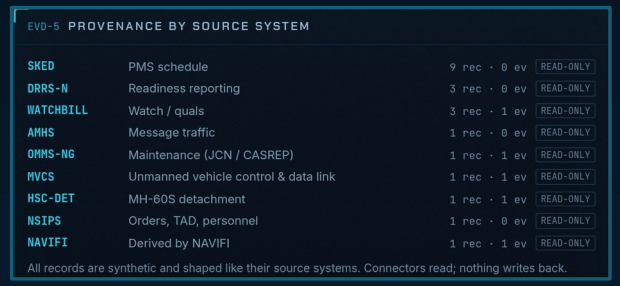
→ **Change event**
Time-stamped, with its source record attached

→ **Mission recomputed**
in milliseconds, every number linked to its record

Connectors only read. HAWSIR never writes back to a system of record.

Bad data gets flagged, not retyped

Missing	plans on a conservative value and says so
Stale	flagged on the input that controls
Contradictory	plans on the worse value, shows both
Malformed	rejected; the last good value is kept


EVD-5 PROVENANCE BY SOURCE SYSTEM

SKED	PMS schedule	9 rec · 0 ev	READ-ONLY
DRRS-N	Readiness reporting	3 rec · 0 ev	READ-ONLY
WATCHBILL	Watch / quals	3 rec · 1 ev	READ-ONLY
AMHS	Message traffic	1 rec · 0 ev	READ-ONLY
OMMS-NG	Maintenance (JCN / CASREP)	1 rec · 1 ev	READ-ONLY
MVCS	Unmanned vehicle control & data link	1 rec · 1 ev	READ-ONLY
HSC-DET	MH-60S detachment	1 rec · 1 ev	READ-ONLY
NSIPS	Orders, TAD, personnel	1 rec · 0 ev	READ-ONLY
NAVIFI	Derived by NAVIFI	1 rec · 1 ev	READ-ONLY

All records are synthetic and shaped like their source systems. Connectors read; nothing writes back.

The provenance panel: how many records each system supplied, all read-only.

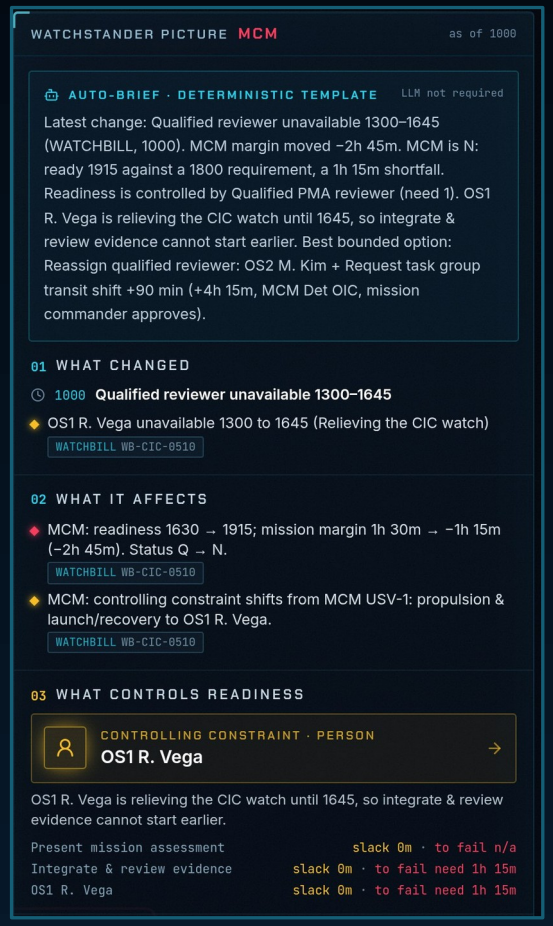
The paperwork comes from the code

- SBOM and signature on every image build
- 48 NIST 800-53 controls in an OSCAL file generated from the code
- CI fails if that file drifts from the code

THE SAILOR AT THE CONSOLE

Built for the person on watch

One screen, plain words, and it keeps working when the ship goes quiet.



The written brief on the watch floor

1 Five answers, always in this order

Is it up now? Will it be up when needed? What controls it? How much margin? What are the options?

2 Plain words first

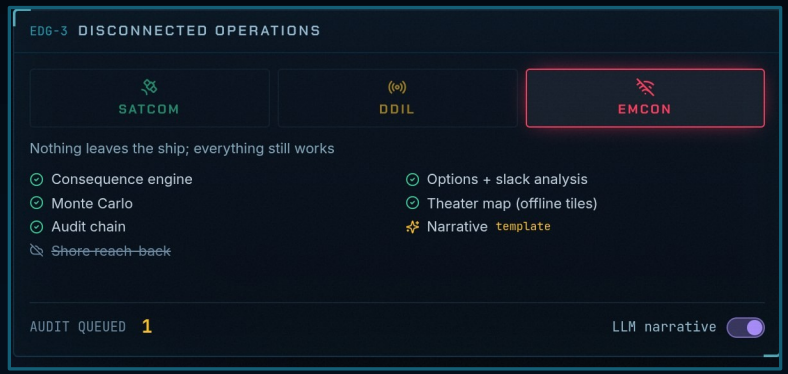
A written brief on top, numbers underneath. AI text is violet and labelled; with no model it falls back to a template.

3 One button to stop it

STOP halts every recommendation at once, and the stop itself is logged.

4 Hands stay on the keyboard

1–9 jump between screens, Space plays the change stream, Shift+R resets the session.



Flip to EMCON: nothing leaves the ship, and everything still works.

What we haven't done yet

Put it in front of a real watch team. That is the first thing in the 30-day plan.

WHO DECIDES

The AI explains. A person decides.

We kept the AI out of the math on purpose.

Deterministic engine

Same inputs, same answer, every time.

The AI only narrates

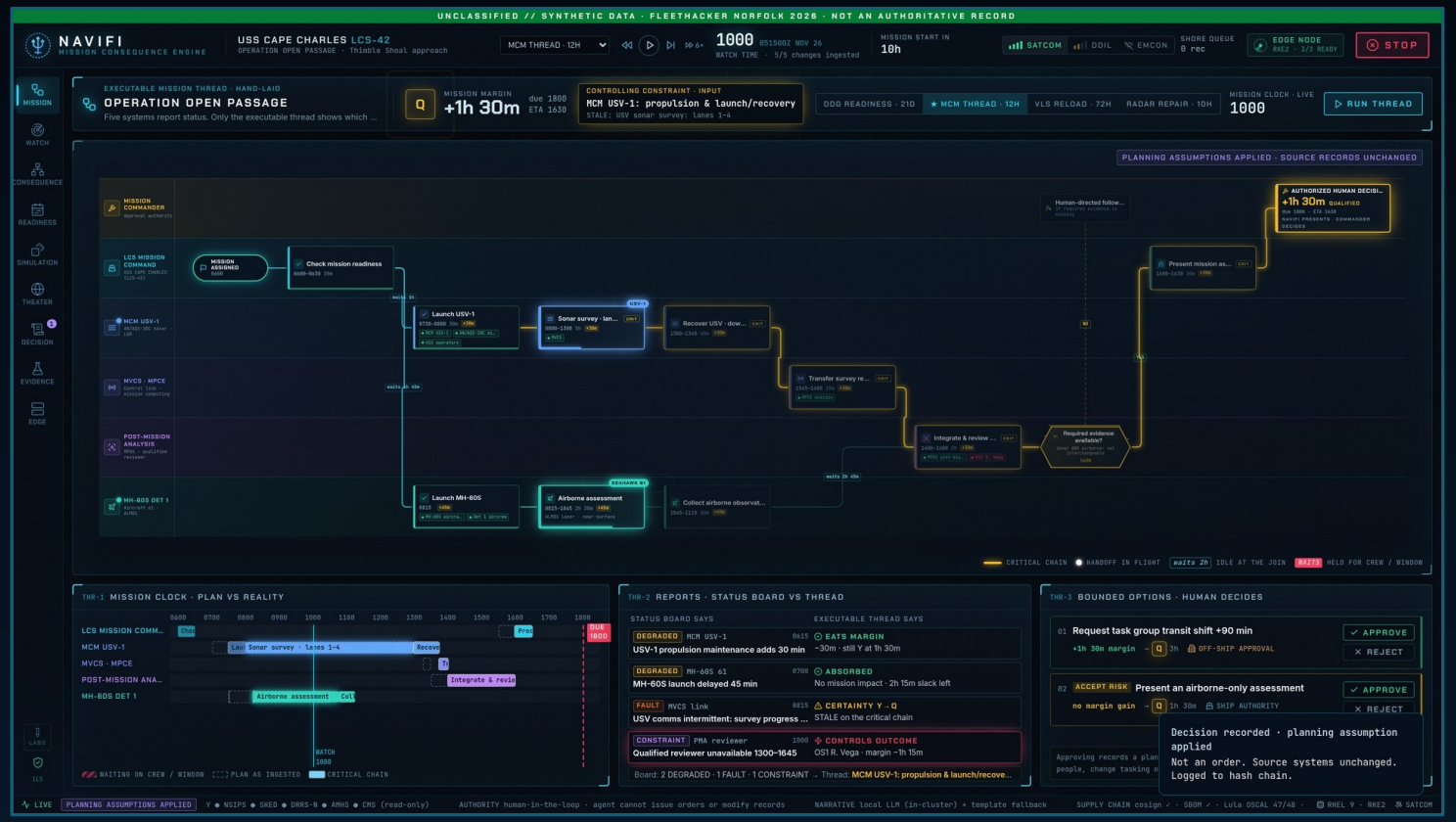
Switch the language model off and the answer is identical.

Approve, reject or STOP

Approving applies a planning assumption, never an order. Source records stay untouched.

Every decision on a hash chain

Tamper with the log and verification catches it, then restores from the edge copy.



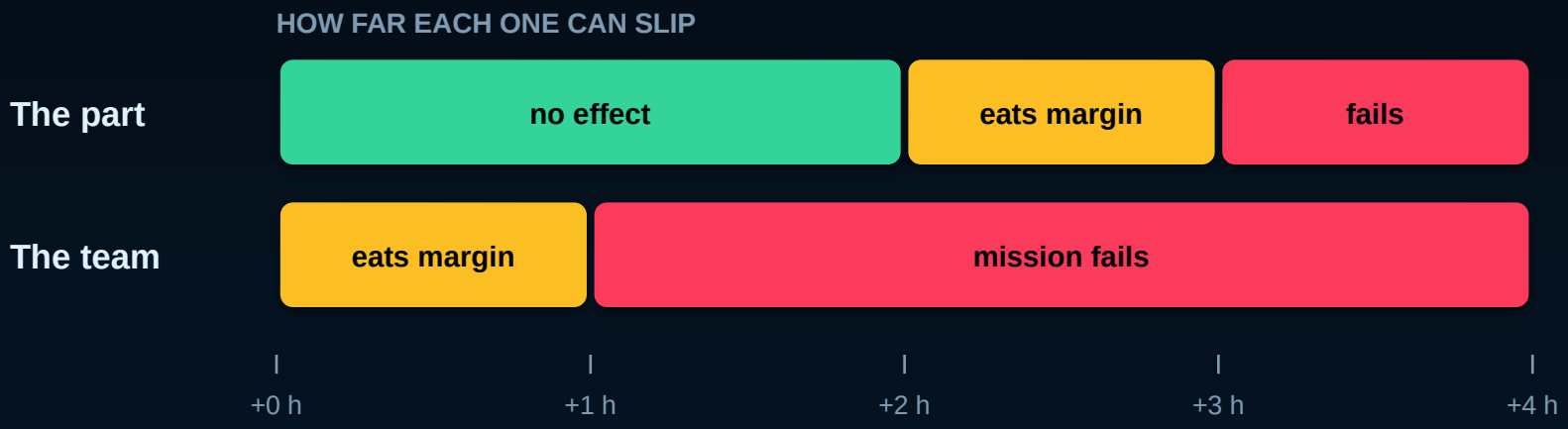
After a person approved one option: −1h 15m became +1h 30m, and the decision went on the log

THE IDEA UNDER THE ENGINE

Slack is not margin

The late part gets the attention. The input that controls the mission often isn't the late one.

A radar repair: part due 1200, qualified team free at 1400, a 3-hour job, needed by 1800.
Ready at 1700 with 1 hour of margin. The team controls, not the part.



Slack

How far an input slips before anything changes. The part: 2 hours. The team: none.

Margin

How far the mission can slip before it fails. Here 1 hour, and the team sets it.

Exact, not estimated

Readiness rules only move one way, so bisection finds both numbers exactly for every input.



The same check in the app: OS1 Vega on the mine countermeasures thread

THEN WE TRIED TO BREAK IT

Twelve ways a real watch could trip it up

We wrote the failure cases down, and the app runs every one of them live against the engine.

Bad data

- Missing input
- Stale input
- Contradictory sources
- Malformed record

Bad conditions

- Disconnected (EMCON)
- LLM unavailable

Hard problems

- Nonlinear consequence
- Hidden cliff
- No single fix works

Known answers

- Classic baseline
- Part slips to 1330
- Team slips to 1600



EVD-2 FAILURE CASES · RUN LIVE AGAINST THE ENGINE — 12/12 PASS — ▷ RUN ALL

CASE	INJECTED	EXPECTED	OBSERVED (LIVE)	RESULT	MS
Classic: baseline	Part 1200 · team 1400 · repair+test 3h · required 1800	Ready 1700, +1h margin, team controls	Ready 1700 · +1h · Qualified repair team available	✓ PASS	0.2
Classic: part slips to 1330	Part ETA 1200 → 1330	No mission consequence (upstream slack absorbs it)	Ready 1700 · margin still +1h	✓ PASS	0.1
Classic: team slips to 1600	Team availability 1400 → 1600	Misses 1800 by 1h, NOT READY	Ready 1900 · -1h · N	✓ PASS	0.0
Missing input	Part ETA deleted from the feed	Flag MISSING, use conservative assumption, never silently READY	Flagged MISSING · assumed 1500 · Q at +0m	✓ PASS	0.0
Stale input	Team status last reported 0800; now 1100, staleness limit 2h	Flag STALE on the controlling input	Flagged STALE on repair team · status Q	✓ PASS	0.0
Contradictory sources	Watchbill says team 1400; AMHS message says 1500	Flag CONTRADICTORY, plan on the worse value	Flagged CONTRADICTORY · planned on 1500 → ready 1800	✓ PASS	0.0
Malformed record	Supply record arrives with date "2026-13-45T25:99"	Reject record, flag MALFORMED, keep last good value	Flagged MALFORMED · last good 1200 kept · no engine errors	✓ PASS	0.1
Disconnected (EMCON)	All network calls fail during the recompute	Full answer computed aboard, zero network calls	6 capabilities + options computed · 0 network calls	✓ PASS	5.2
LLM unavailable	Narrative model offline	Identical calculated answer; template explanation	Status, margin and control identical · LLM never in the calculation path	✓ PASS	0.4
Nonlinear consequence	CFA ETA slips 2d 3h (past the sail date)	Detect the missed Norfolk window; Rota controls; margin loss ≫ slip	2d 3h slip → -9d 18h margin · Rota port visit	✓ PASS	0.3
Hidden cliff	Helo detachment fly-on, currently on time	Warn that a small slip misses the fly-on window	Cliff after 2h slip · drop 5d 14h	✓ PASS	1.7

The failure-case table in the app, all twelve passing

We wrote down what it does not do yet, too

Durations are planner estimates, delays are treated as independent, and the picture is one ship. Those limits are on the evidence screen next to the results.

12 failure cases, run live

39 app tests

48 tests on Connor's engine

ONE ENGINE

Then we pointed it at three more missions

We laid out the MCM thread by hand. The same engine laid out the rest from their dependency graphs.

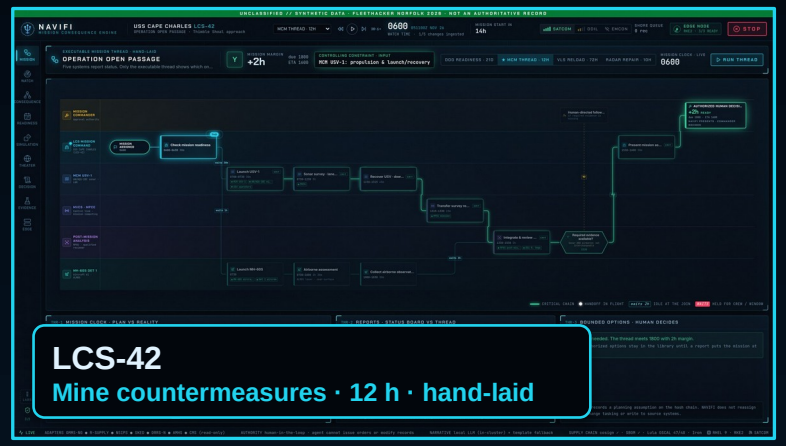
4 mission threads

2 ship classes

3 hulls

0 engine rewrites

Threads, interfaces and deployment profiles are configuration, so a new ship class is data work, not code.





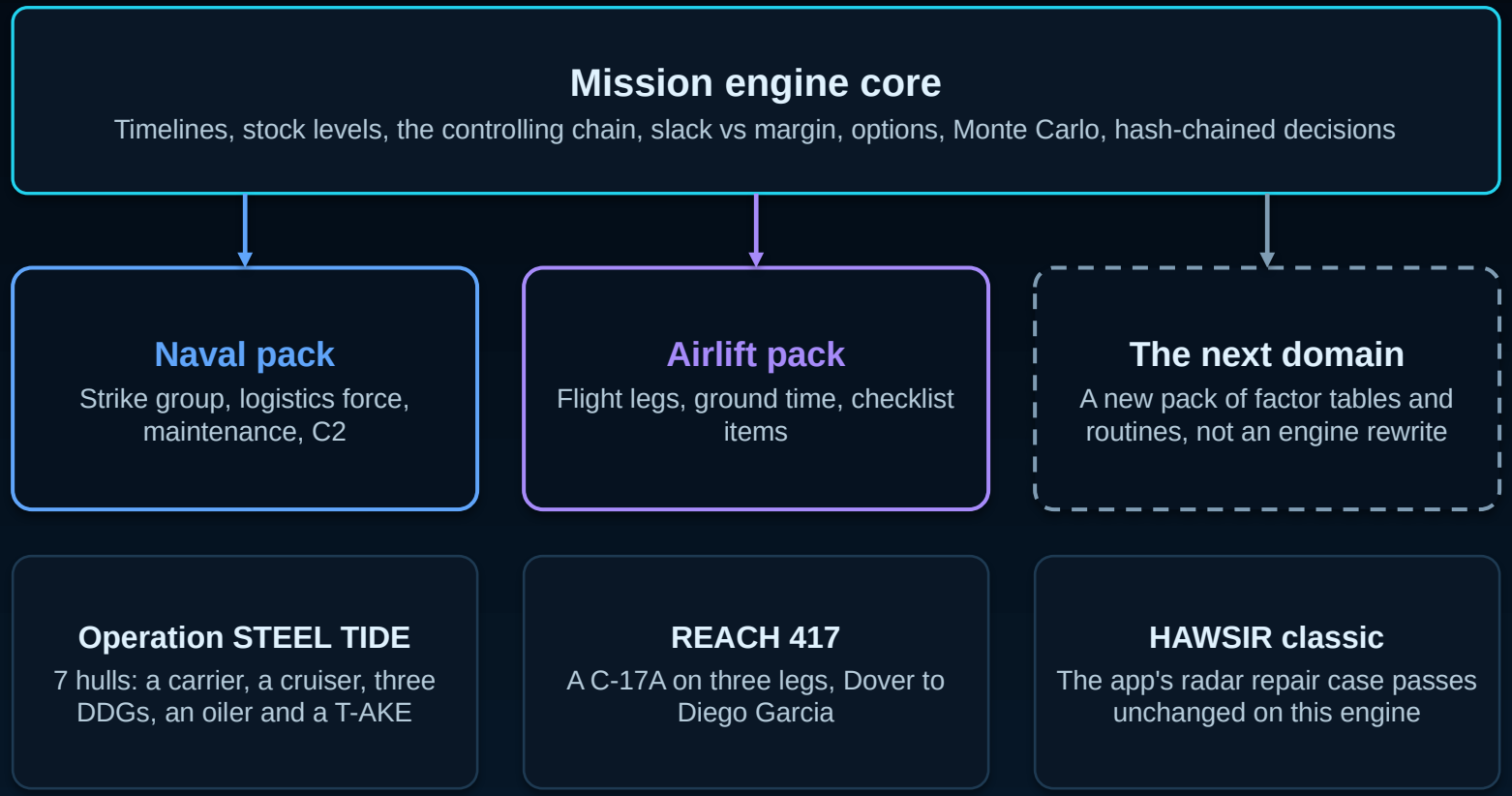




OTHER SHIPS, OTHER DOMAINS

Beyond the surface fleet

Connor's engine keeps the core general and puts the domain knowledge in packs.



- Packs work together: the naval pack calls the airlift pack to check the carrier onboard delivery flight.
- Crews are tracked by qualification (AMD, strike, FC, ET), so who can cover a task is part of the math.

In the HAWSIR app today

- Four mission threads on one engine
- Two ship classes, three hulls
- A fleet picture prototype: 38 hulls, 12 missions, 140+ packages timed by the engine

Not yet

The app still runs its own engine. Wiring Connor's in is the next job; his engine's 48 tests run on their own today.

HAWSIR · INFRASTRUCTURE

Built this week: a full Kubernetes platform on AWS

From terraform apply to a hardened cluster serving HAWSIR at **navifi.uds.demo.uncomn.ai**, all from code in the repo

1 TERRAFORM
Provision AWS
Node, firewall, IAM role, load balancer rules, TLS certificate and DNS

27 resources, one apply

→

2 FIRST BOOT
Harden the host
RHEL 9 with SELinux enforcing. SSM agent instead of SSH, no key pair

user-data.sh

→

3 RKE2
Install Kubernetes
Single-node RKE2 v1.36 with SELinux, plus the kernel setup Istio needs

install-rke2.sh

→

4 UDS CORE
Install the platform
Zarf init, then UDS Core 1.14: policy, mTLS, SSO, logs, metrics, backups

install-core.sh, signed

→

5 HAWSIR
Deploy the app
Any git commit: build, test, package with Zarf, roll out into the cluster

deploy.sh, SBOM per build

Where it runs

AWS us-east-1, inside the existing lab VPC

EC2 m7i.2xlarge: 8 vCPU, 32 GiB

RHEL 9 with SELinux enforcing

Behind an AWS load balancer with TLS

navifi.uds.demo.uncomn.ai

Real problems we solved on the way

SELinux locked a 2nd registry pod out of its volume
→ pinned the registry to one replica

RKE2 1.36's Gateway API CRDs blocked UDS Core
→ turned RKE2's bundled ingress off

MinIO stopped publishing public images
→ moved to the Chainguard build

Tight storage rules cut Loki off from the API
→ added explicit Kubernetes API rules

All in code

27 AWS resources from one Terraform apply

4 scripts from first boot to live app

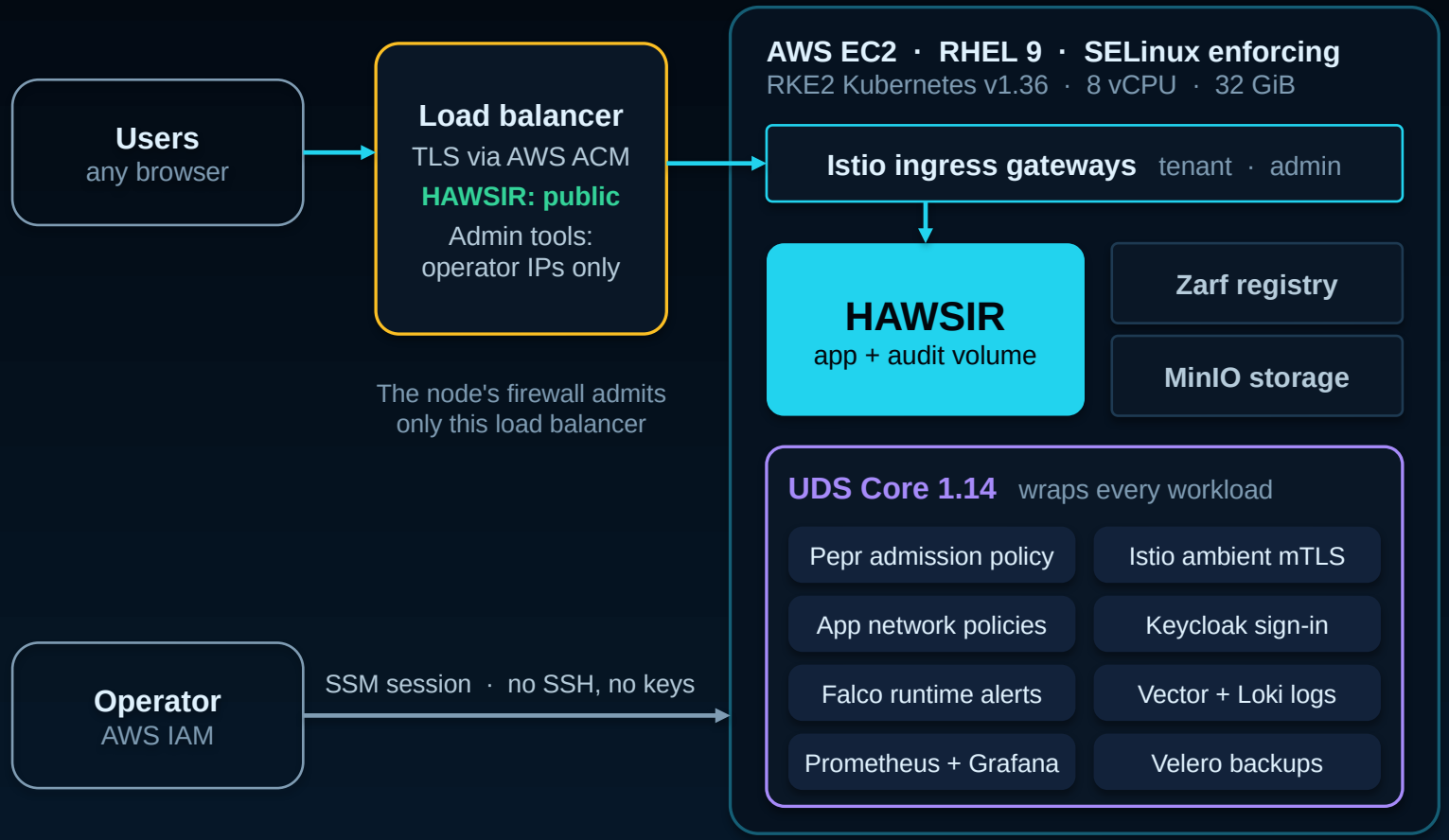
3 Zarf and UDS packages, platform to app

Rebuild it from infra/README.md

HAWSIR · LAB CLUSTER ON AWS

One hardened node running the ship's stack

RHEL 9 · RKE2 Kubernetes · Defense Unicorns UDS Core · Zarf live at **navifi.uds.demo.uncomn.ai**



Locked down by default

- SSM sessions only, logged in CloudTrail
- Only HAWSIR is public; admin tools return 404
- Default-deny network policy and mTLS

Commit to live in one command

- Typecheck and tests gate every image build
- Zarf packages it with an SBOM, air-gap style
- Every commit gets its own image tag

Mirrors the shipboard deploy

- Same RHEL 9, RKE2, UDS Core and Zarf
- Lab only: TLS at the load balancer, sign-in off
- Ship adds DISA STIG, FIPS and Keycloak SSO

8 vCPU · 32 GiB
the whole platform on one node

0
node ports open to the internet

Signed
Sigstore-verified, SBOM per build

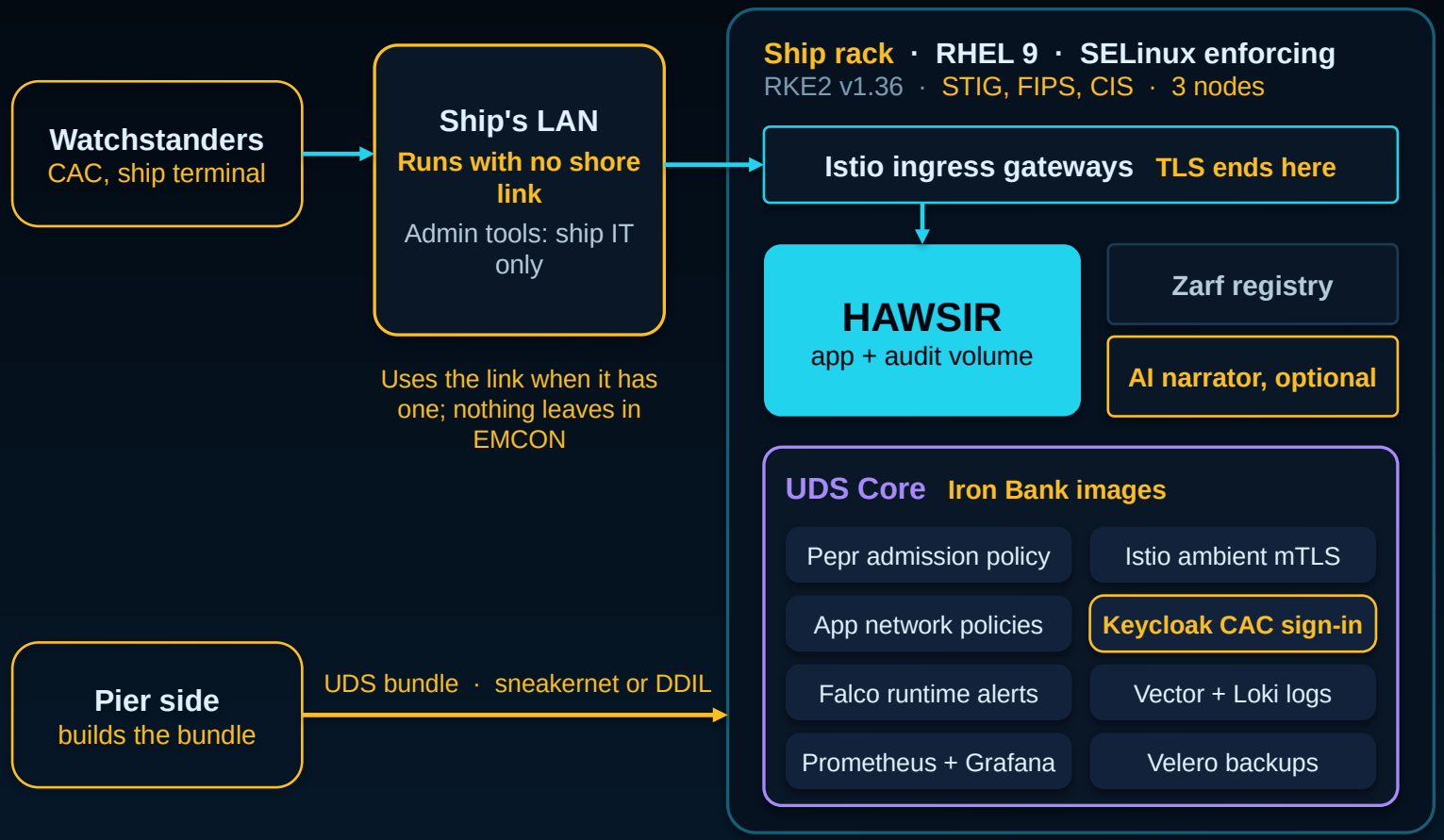
1 command
from any git commit to live

HAWSIR · THEORETICAL SHIPBOARD DEPLOYMENT

The same stack aboard, built for DDIL

RHEL 9 · RKE2 Kubernetes · Defense Unicorns UDS Core · Zarf same as the lab

Amber: changes from the lab



Locked down aboard

- Installed whole; nothing downloads at sea
- CAC sign-in through Keycloak
- Default-deny network policy and mTLS

Works with the link down

- SATCOM: audit goes ashore in batches
- DDIL: one entry at a time, local model
- EMCON: nothing leaves, everything works

What changes from the lab

- One air-gapped bundle, not a git build
- TLS at the Istio gateway, no load balancer
- DISA STIG, FIPS, CIS and Iron Bank images

1 bundle
Zarf init, UDS Core and HAWSIR

0
downloads at sea, model included

Signed
Sigstore-verified, SBOM per build

3 link states
SATCOM, DDIL and EMCON

WHERE WE ARE

What's real, what isn't yet, and what's next

Three days in. Here is the straight version.

Real today

- Runs on the ship's stack, live on AWS
- Four mission threads on one engine
- Twelve failure cases and a hash-chained decision log
- Any git commit to live in one command

Not real yet

- All data is synthetic
- Durations are planner estimates, not fleet history
- Correlated delays (one cause, many slips) aren't modeled
- One ship at a time; squadron roll-up is a prototype

Next

- Wire Connor's new mission engine into the app
- 30 days: read-only exports from one DDG; three watch teams try it against today's process
- 60 days: workflow authoring, and UDS on an RKE2 edge node (we did a version of this one this week)
- 90 days: three hulls plus a squadron roll-up

Synthetic data only. A human decides.

APPENDIX

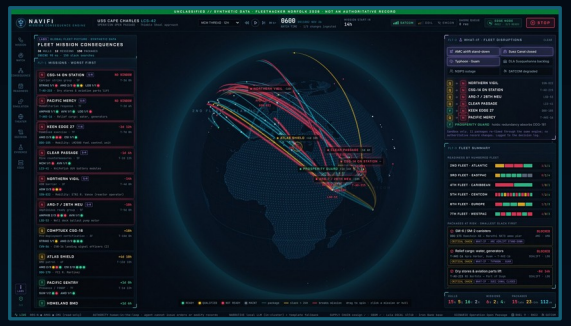
The app, screen by screen

Screenshots from the running app, for anyone who wants to look closer. All scenario data is synthetic.



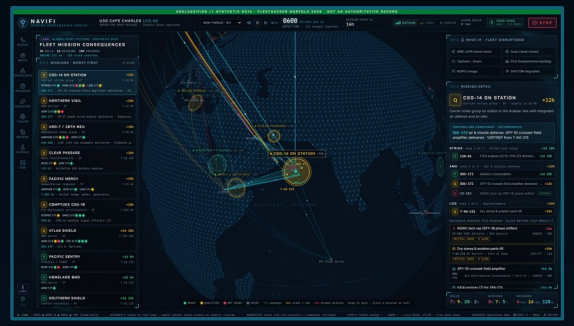
FLEET PICTURE

The whole fleet on one globe



FLEET WHAT-IF

Three disruptions at once



FLEET DRILL-DOWN

One mission, hull by hull



MONTE CARLO

2,000 seeded futures



MONTE CARLO

The radar repair, run 2,000 times



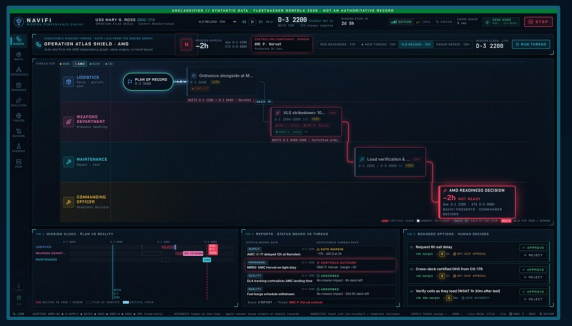
READINESS HORIZON

Deadlines before they hit



THEATER

Where the parts and people are



OTHER WORKFLOWS

Four threads, one engine

APPENDIX · FLEET PICTURE

The whole fleet on one globe

A labs prototype: every mission, hull and inbound package, timed by the same engine.



Fleet picture at rest, synthetic data. Drag to spin; click a mission or hull to drill in.

1 38 hulls, 150 packages

Twelve missions across six numbered fleets. Each mission ring takes its readiness color, and the list on the left runs worst first.

2 Every arc is a package

Parts, people and data moving toward a hull. Amber has under a day of slack; red breaks its mission.

3 Slack before it breaks

The right panel lists packages at risk, smallest slack first, and marks the ones on the critical chain.

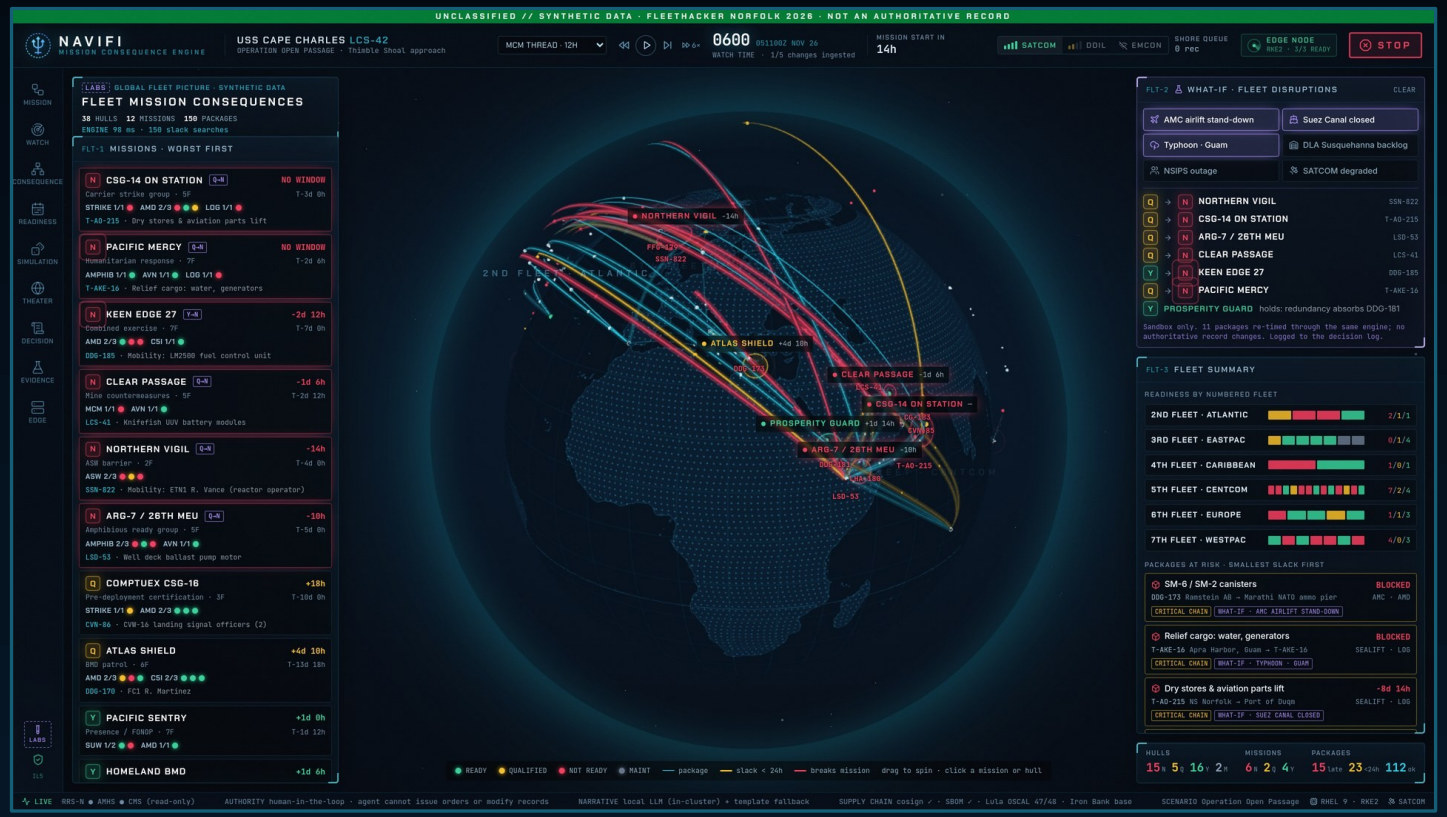
4 Fast enough to explore

The engine re-times every package in a fraction of a second on a laptop, so a what-if answers instantly.

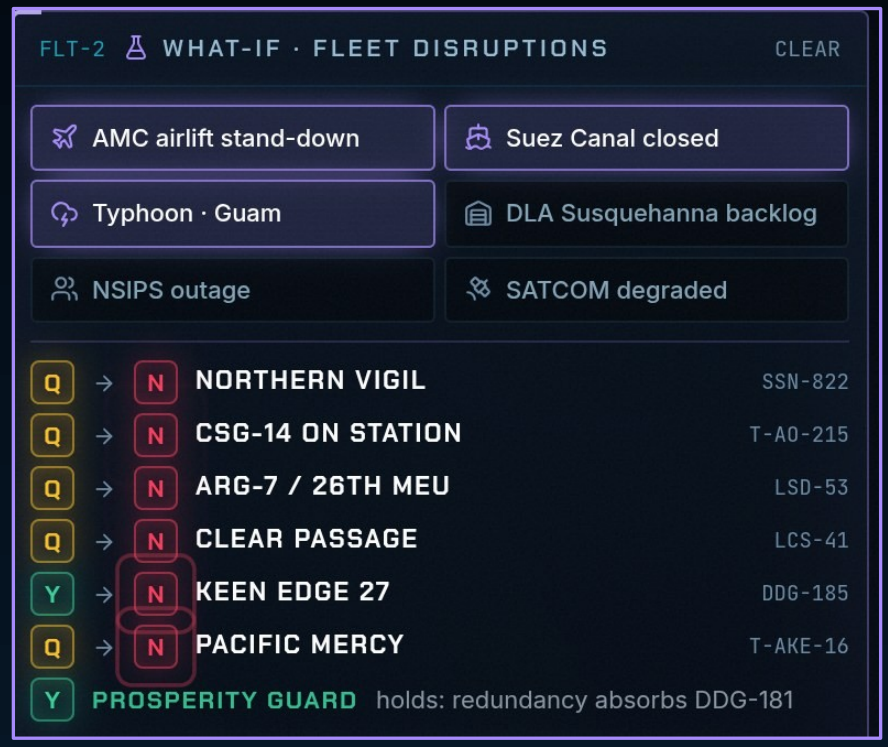
APPENDIX · FLEET WHAT-IF

What if three things go wrong at once?

Toggle a disruption and see which missions break, and which ones redundancy saves.



AMC stand-down, Suez closed and a typhoon at Guam, all at once



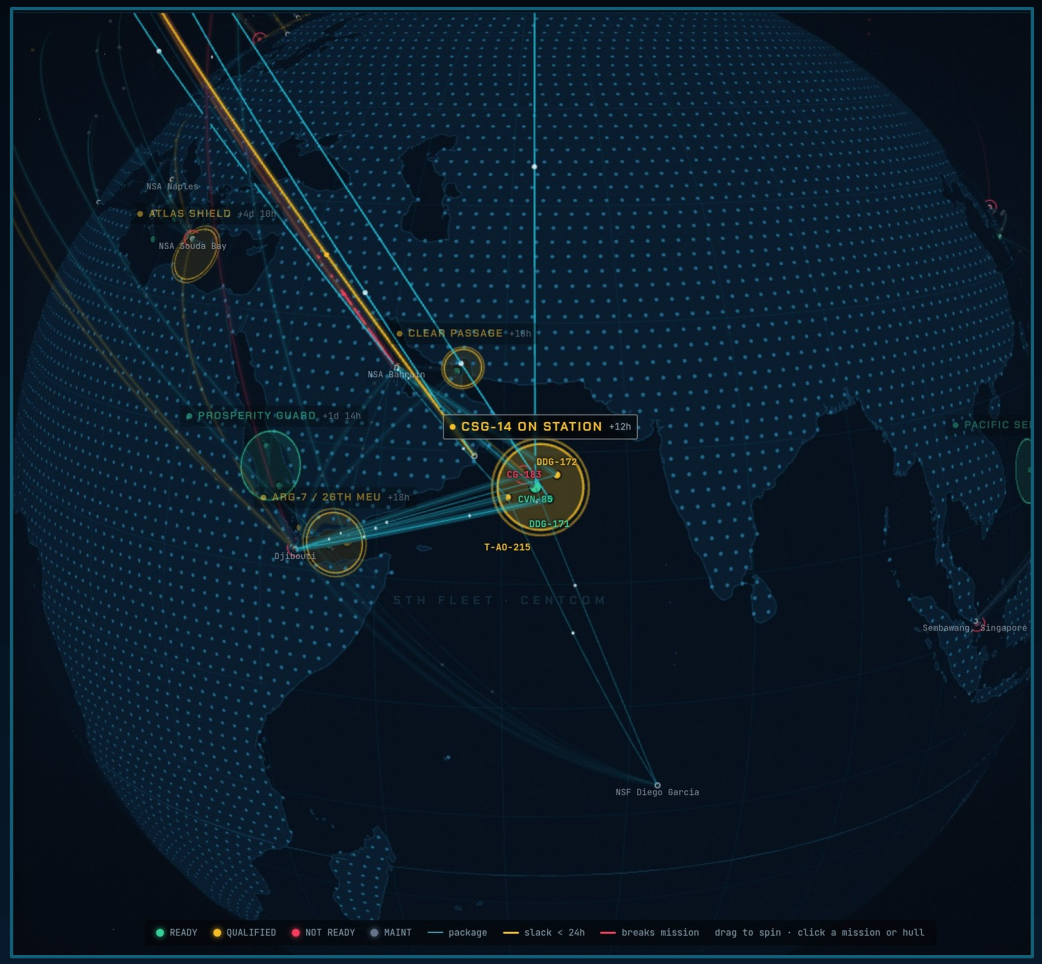
The what-if panel, zoomed in

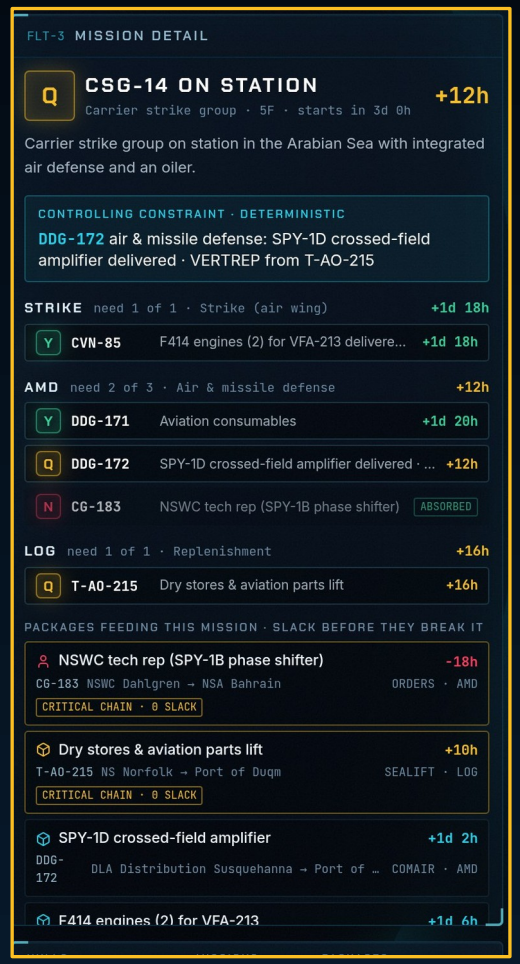
Six missions drop to N. PROSPERITY GUARD holds: it has more hulls than it needs, so DDG-181 running late is absorbed. Sandbox only; no record changes, and the what-if is logged.

APPENDIX · FLEET DRILL-DOWN

One mission, hull by hull

CSG-14 on station in the Arabian Sea: what it needs, who supplies it, and what controls it.





1 **Need 2 of 3, not 3 of 3**

Air defense needs 2 of 3 ships. CG-183 is N, but the other two cover, so it is absorbed.

2 **One controlling constraint**

DDG-172 waiting on a SPY-1D amplifier by VERTREP from the oiler sets this mission at +12 h.

3 **Slack on every package**

Each inbound part and person shows how far it can slip before it breaks the mission.

4 **Down to the hull**

Click a hull for its capabilities and packages. Two hulls open their full watch floor.

APPENDIX · MONTE CARLO

2,000 seeded futures, not one guess

USS Grace Hopper, 14 days out. Every uncertain input is drawn from a range and the whole graph re-runs.



Simulation screen, DDG readiness thread, synthetic data

1 Readiness S-curves

The chance each capability is ready by any time. Five are solid; air and missile defense is the one at risk.

2 Controls across futures

FC1 Martinez controls air defense in most futures; the CFA part in about one in five.

3 Seeded and reproducible

Same seed, same answer, so the watch and a reviewer see identical results. Thousands of futures a second, in the browser.

4 Honest about its inputs

Ranges are planning assumptions widened for stale and missing data, not calibrated failure rates. The screen says so.

APPENDIX · MONTE CARLO

The radar repair, run 2,000 times

The same case as 'slack is not margin'. Across every future, the team controls and the part never does.



Simulation screen, radar repair thread, synthetic data

1 Margin is a spread

Most futures make 1800 with time to spare, and a real tail misses. The histogram shows how much and how often.

2 The team always controls

The qualified repair team sets readiness in all 2,000 runs. The replacement part sets it in none.

3 What moves the answer

The tornado pushes each input to the ends of its range. Only the team and the repair time move the margin.

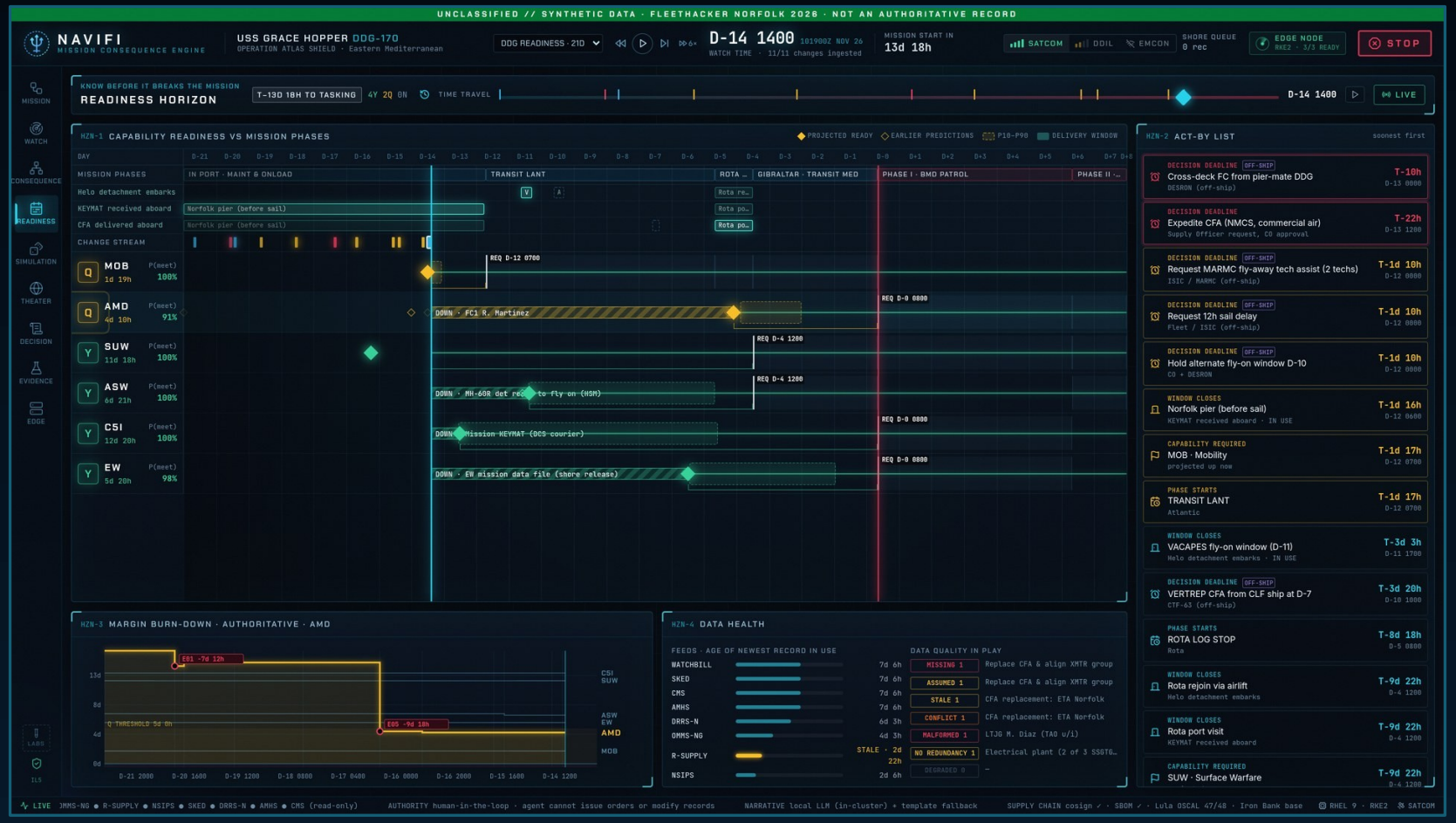
4 One hour, tested

The deterministic answer is an hour of margin. The simulation shows how often that hour actually holds.

APPENDIX · READINESS HORIZON

Deadlines before they hit

Every capability against the voyage phases, with the next decisions in the order they come due.



1 Readiness vs phases

Each capability gets a projected-ready mark, its earlier predictions and a P10–P90 band over the voyage.

2 Act-by list

Soonest first: when a fly-on window closes, when an off-ship request has to go out.

3 Margin burn-down

How margin has moved as reports landed, against the line where Y drops to Q.

4 Data health

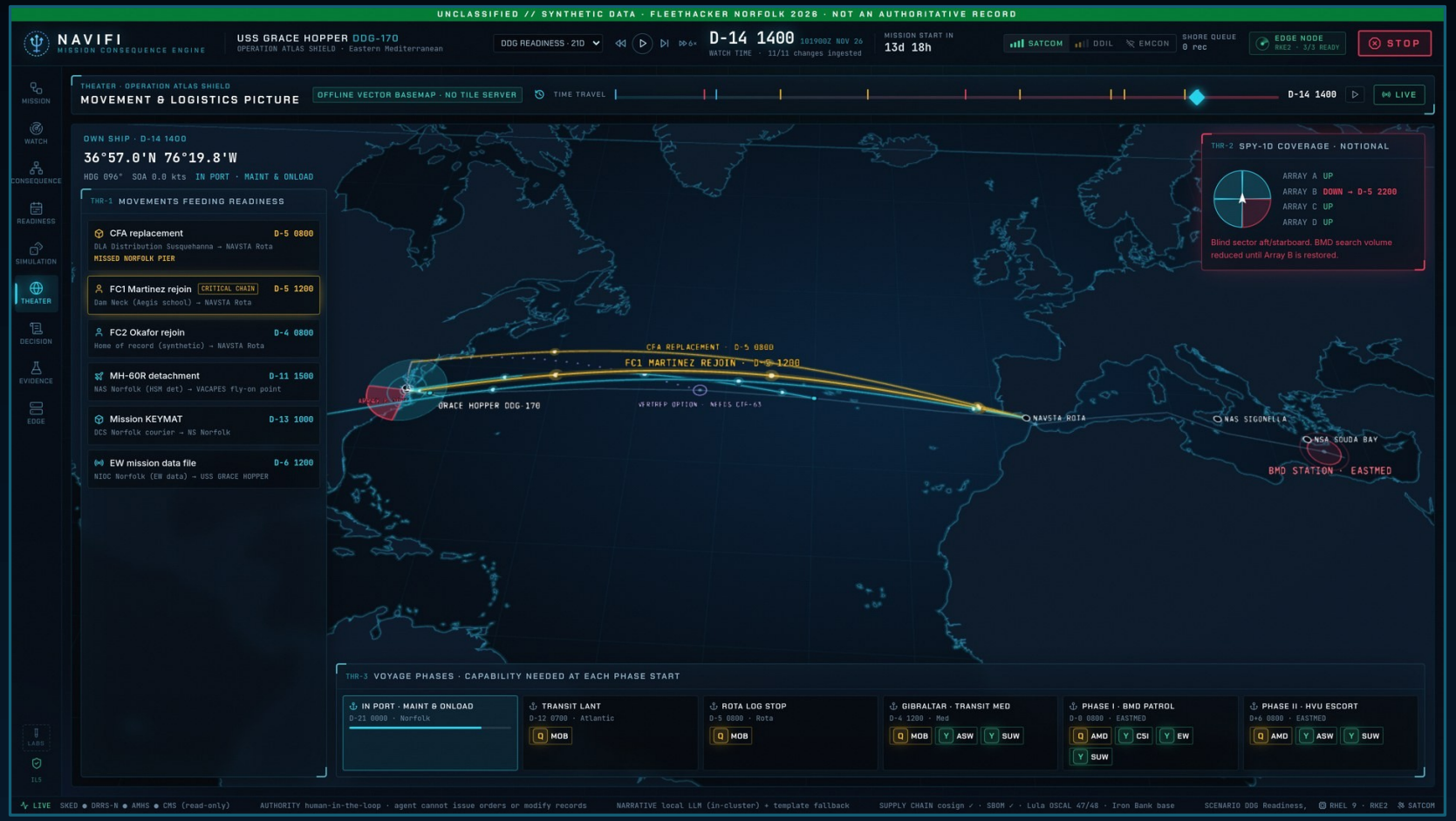
The age of the newest record from each feed, and every stale, assumed, conflicting or missing value.

Readiness horizon, DDG readiness thread, synthetic data

APPENDIX · THEATER

Where the parts and people are

Grace Hopper in Norfolk, 14 days before a BMD patrol in the eastern Med.



Theater map, DDG readiness thread, synthetic data

1 Movements that matter

Every part, person and data file en route, with its ETA. FC1 Martinez rejoining from Dam Neck is the critical chain.

2 Phases need capabilities

Each voyage phase lists what it needs at its start, so a late part shows up as a phase at risk.

3 Radar coverage

SPY-1D Array B is down until D-5, and the map shows the blind sector it leaves. Notional.

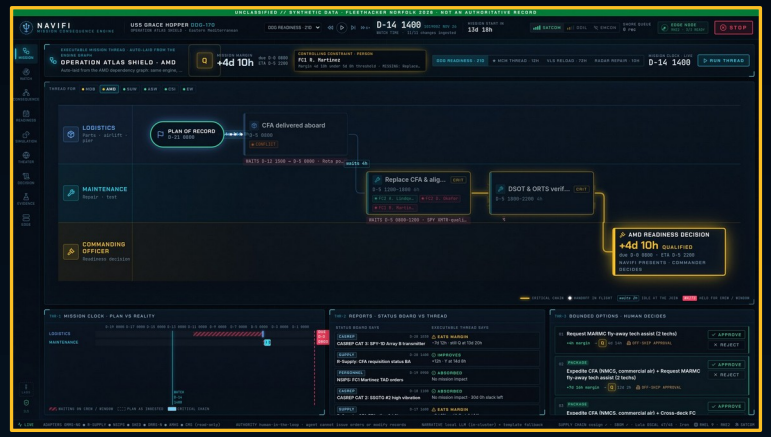
4 Works in EMCON

An offline vector basemap with no tile server, so the picture holds with the link down.

APPENDIX · OTHER WORKFLOWS

Four mission threads, one engine

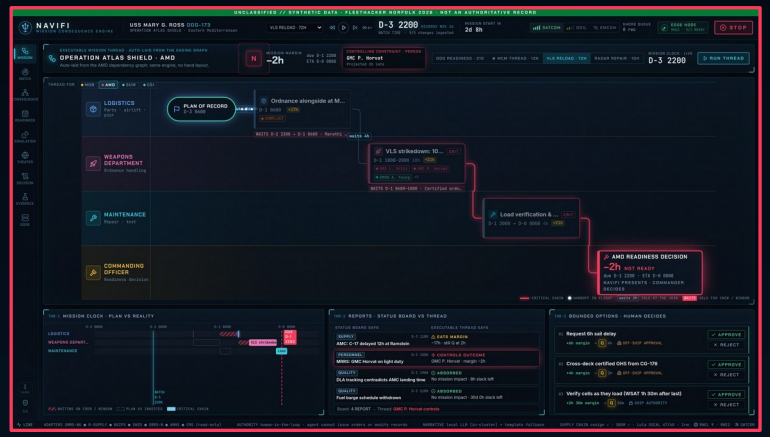
The same engine and the same screens, from hours to weeks and from parts to people.



DDG READINESS · 21 DAYS

Grace Hopper before deployment

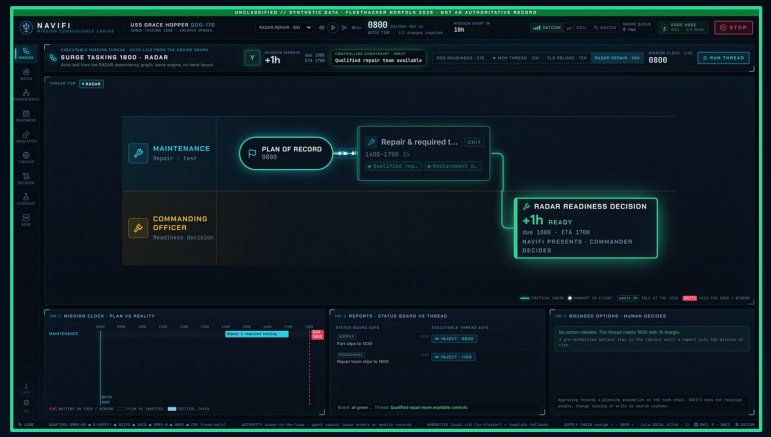
Air defense is Q at +4 d 10 h. FC1 Martinez, not the late part, controls it.



VLS RELOAD · 72 HOURS

Mary G. Ross at Souda Bay

Air defense is N at −2 h. GMC Horvat on light duty controls the reload, not the delayed C-17.



RADAR REPAIR · 10 HOURS

Grace Hopper surge tasking

Y at +1 h. The qualified team controls; the part has two hours of slack.

The fourth is in the main deck: mine countermeasures on Cape Charles, 12 hours. In every one, the thing that controls the mission is not the thing everyone is chasing.